Power Shield:
An Automated Yield-Loss Alerting (AYLA) Pipeline Using Explainable AI (XAI) and Machine Learning (ML) for Power Grid Attack Detection and LLM-Guided Remediation

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Abstract

Power grids face coordinated cyber-physical attacks—including False Data Injection (FDIA), Relay Saturation, and Remote Terminal Communication Injection—designed to evade conventional anomaly detectors by mimicking plausible physical behavior. This paper presents **Power Shield**, implementing an **Automated Yield-Loss Alerting (AYLA)** pipeline: a four-stage explainable AI system that detects grid attacks, attributes each alert to specific sensor drivers via SHAP, retrieves the matching remediation protocol via RAG, and delivers step-by-step operator guidance via a local LLM. The SHAP-guided stacked ensemble achieves **91.88%** accuracy and **94.6%** attack recall on the full combined MSU/ORNL SCADA dataset (78,377 samples), surpassing the established combined-dataset baseline. Stages 2–4 (SHAP, RAG, and LLM) activate only on classifier-flagged events, not on every sensor reading, so the expensive LLM inference is incurred only when explanation and remediation are genuinely needed. A no-context ablation confirms LLM grounding: withholding retrieved protocol documents drops attribution accuracy from **90.5%** to **16.5%** (−74.0 pp, non-overlapping 95% CIs), proving that pipeline accuracy reflects genuine document grounding rather than parametric knowledge.

1 Introduction

Power grids are among the most critical infrastructure systems in modern society. State-sponsored attacks—including the Ukrainian blackouts of 2015–2016 [1]—demonstrated that adversaries can weaponize SCADA control software with devastating physical consequences. ICS attacks increased by over 140% between 2020 and 2024.

Traditional SCADA intrusion detection relies on rule-based signature matching that fails against False Data Injection Attacks (FDIA), which craft a sensor manipulation vector that passes the bad-data detection residual test [2]. Machine learning classifiers have emerged as a complement to signature-based systems, learning multi-variate anomaly patterns directly from sensor data. However, detection alone is insufficient: an operator who receives an alert without knowing *which* sensors triggered it, *which* attack scenario it matches, and *what* corrective steps to take is at best slowed and at worst misled. Current commercial ICS/OT platforms (Claroty, Dragos, Nozomi Networks) surface alerts but do not generate automated, grounded, step-by-step remediation guidance.

A critical methodological distinction separates prior literature on the MSU/ORNL power grid dataset: prior work achieving high detection accuracy [3–5] evaluates classifiers on individual attack sub-scenarios *in isolation*, eliminating the inter-class ambiguity that characterises combined-dataset deployment. The only prior work applying explainability to the *combined* MSU/ORNL dataset [6] demonstrates SHAP feature attribution but does not extend to automated protocol retrieval or operator remediation.

Power Shield implements AYLA to close this gap end-to-end. Its three principal contributions are:

- **C1. SHAP-guided RAG query construction.** Per-event SHAP attributions identify the top contributing sensors and enrich their names with semantic type descriptors (e.g., `R1-PA:ZH` → `R1-PA:ZH (impedance harmonic)`) before forming the RAG retrieval query, directly linking the classifier's explanation layer to document retrieval. Prior work uses SHAP for post-hoc feature explanation [6,7]; AYLA uses per-event attributions as *semantic retrieval keys*—to the author's knowledge, a novel application.
- **C2. Ablation-confirmed LLM grounding (XAI transparency verification).** A no-context ablation withholds retrieved protocol documents while keeping all other prompt inputs (SHAP drivers, sensor readings, classifier verdict) identical. Attribution

accuracy drops from 90.5% to 16.5%—a 74.0 pp decline with non-overlapping 95% CIs—directly confirming that full-pipeline accuracy reflects genuine document grounding, not parametric knowledge. This ablation is an XAI-style transparency verification: it proves empirically that the pipeline's recommendations trace to specific retrieved protocol documents rather than opaque parametric memory, establishing end-to-end decision traceability from sensor reading through SHAP attribution, retrieval, and LLM reasoning.

C3. Combined-dataset evaluation surpassing the baseline. AYLA operates on the full heterogeneous MSU/ORNL dataset and surpasses the Hink et al. [8] combined-dataset baseline (90.56%)—the only prior result directly comparable to a combined-dataset formulation—while adding SHAP attribution, RAG protocol retrieval, and LLM-guided remediation.

2 System Architecture

The AYLA pipeline processes each grid event through four sequential stages. Stages 1–2 operate on raw sensor data; Stages 3–4 operate on the symbolic output of Stage 2. Critically, Stages 2–4 (SHAP attribution, RAG retrieval, and LLM remediation) activate only when Stage 1 exceeds τ: the classifier runs at machine speed on every incoming sensor snapshot, while LLM inference is reserved for the small fraction of events that require explanation and remediation. This event-driven design makes the pipeline viable for real-time SCADA deployment without high-throughput inference infrastructure.

Stage 1 — Probabilistic Classification. A LightGBM/XGBoost stacked ensemble trained on 29 SHAP-selected features (from an initial pool of 128) ingests a sensor snapshot and outputs $P(\text{Attack})$ via a LogisticRegression meta-learner. If $P(\text{Attack}) > \tau = 0.560$, the event is flagged and passed to Stage 2.

Stage 2 — SHAP Attribution. TreeSHAP [9] computes exact SHAP values for all 128 features via the LightGBM base model. The top-k positive contributors (sensors pushing toward Attack) are selected as *alert drivers* and expanded to semantic descriptors using a relay-and-type regex registry. SCADA, relay, and Snort IDS log channels are reported independently from physical sensor drivers.

Stage 3 — RAG Protocol Retrieval. A two-phase retrieval strategy queries a ChromaDB vector store indexed from thirteen operator-written grid-protocol manuals using `all-MiniLM-L6-v2` sentence embeddings. *Phase 1* (bi-encoder, multi-pass): a relay-filtered primary query (k=4) surfaces relay-specific FDIA protocols; supplementary type-pattern queries (k=2) surface generic protocols. *Phase 2* (cross-encoder reranking, adopted from IHGR-RAG [10]): all candidates are rescored jointly against the primary query by a `ms-marco-MiniLM-L-6-v2` cross-encoder, producing scores comparable across all retrieval passes regardless of which pass found each candidate.

Stage 4 — LLM Remediation. The ranked protocol context, SHAP summary, sensor readings, and classifier verdict are assembled into a structured prompt. A Chain-of-Thought (CoT) [11] step instructs the LLM to explicitly verify the RANK #1 protocol's diagnostic signature against the observed sensors before committing to a diagnosis. The model outputs: (1) Protocol Match Reasoning, (2) Diagnosis with protocol ID, and (3) numbered remediation steps drawn directly from the matched manual.

3 Mathematical Formulation

SHAP-guided feature selection. An initial LightGBM model trained on all d=128 features computes global mean absolute SHAP importance on a 3000-sample hold-out. Features whose mean absolute SHAP value meets or exceeds the mean across all 128 features are retained, reducing d=128 to d'=29 features.

Ensemble stacking. A LogisticRegression meta-learner combines LightGBM and XGBoost attack probabilities on a held-out calibration set:

$$P(\text{Attack} \mid \mathbf{x}) = \sigma(\beta_0 + \beta_1 P_{\text{LGB}}(\mathbf{x}_S) + \beta_2 P_{\text{XGB}}(\mathbf{x}_S)) \quad (1)$$

where $\mathbf{x}_S$ is the 29-feature SHAP-selected subset, $\beta_1 = 6.74$, $\beta_2 = 0.68$, and $\tau = 0.560$ is selected by sweeping $[0.30, 0.90]$ subject to false alert rate $< 15\%$.

Cross-encoder reranking. For primary query q and candidate document d_j, the cross-encoder f_{CE} scores each pair jointly:

$$\hat{s}_j = f_{\text{CE}}(q, d_j) + f_{\text{CE}}(q_{\text{freq}}, d_j) \quad (2)$$

where q_{freq} is added only when frequency sensors (`:F`, `:DF`) dominate the alert drivers, boosting FREQ-01 retrieval without penalising relay-specific FDIA cases. Unlike bi-encoder L2 scores, cross-encoder scores are directly comparable across all retrieval passes.

4 Dataset and Experimental Setup

4.1 Dataset

Experiments use the MSU/ORNL ICS Security Dataset [14], a widely used power system intrusion detection benchmark generated on a hardware-in-the-loop testbed with a 4-relay transmission protection system communicating via DNP3 and Modbus over a simulated SCADA network. The binary variant contains **78,377 samples** with **128 features**: voltage and current magnitudes, impedance and phase harmonics, relay-level frequency measurements, and SCADA/relay/Snort IDS log channels, spanning four protection relays (R1–R4). The class distribution is 55,663 Attack (71.0%) and 22,714 Natural (29.0%) (2.45:1 imbalance,

Table 1: AYLA system component configuration.

Component	Configuration
Classifier	LightGBM [12] + XGBoost [13] ensemble (29/128 SHAP-selected features)
Meta-learner	LogisticRegression (β_1=6.74, β_2=0.68)
Alert threshold	$\tau = 0.560$ (threshold-sweep optimised; false alert $< 15\%$)
SHAP	TreeSHAP [9] (exact; LightGBM base model; zero-padded to 128-d)
Embedding model	`all-MiniLM-L6-v2` (384-dim); ChromaDB (L2 distance)
RAG Phase 1	Bi-encoder; k=4 relay-filtered + k=2 type-pattern passes
RAG Phase 2	Cross-encoder reranking (`ms-marco-MiniLM-L-6-v2`)
LLM	Mistral-Nemo 12B (Ollama, local; temperature 0.0; max 2048 tokens)

corrected via `is_unbalance=True` and `scale_pos_weight`). Data is split 64/16/20 into training, calibration, and held-out test sets (n=15,676 test samples).

4.2 Protocol Knowledge Base

Thirteen operator-written protocol documents are indexed into ChromaDB using 700-character chunks with 100-character overlap. The protocols span relay-specific FDIA scenarios (R1-FDIA through R4-PM), generic anomaly classes (RSCA-01, RTCI-01, FREQ-01), fault and degradation scenarios (CURR-01, PHASE-01, FAULT-01, DEGR-01, DEGR-02), and a normal-operation baseline (EXPECTED-09).

4.3 Evaluation Protocol

Classifier evaluation uses the full 15,676-sample held-out test set.

RAG retrieval accuracy is assessed on 13 hand-crafted test cases (one per protocol); a case is a hit if the expected protocol identifier appears anywhere in the retrieved context.

LLM attribution accuracy is evaluated on a stratified random sample of 200 triggered events. Ground truth is assigned by matching top SHAP sensors against curated protocol sensor lists via a weighted overlap score; a response is correct if the expected protocol ID appears anywhere in the LLM output.

No-context ablation re-evaluates the same 200 samples with the retrieved protocol context replaced by a placeholder, holding all other prompt inputs constant.

5 Results

5.1 Classifier Performance

Table 2: AYLA ensemble classifier on the held-out 20% test set (n=15,676). CIs are Wilson 95%.

Metric	Value	95% CI
Overall Accuracy	91.88%	[91.44%, 92.30%]
Attack Recall	94.55%	[94.11%, 94.95%]
Natural Recall	85.34%	[84.28%, 86.34%]
False Alert Rate	14.66%	[13.66%, 15.72%]
AUC-ROC / MCC	0.9665	0.8020

The ensemble achieves AUC-ROC of 0.9665 and MCC of 0.8020, indicating strong discriminative power on the full combined dataset despite the 2.45:1 class imbalance. AYLA's 91.88% accuracy surpasses the Hink et al. [8] combined-dataset baseline of 90.56%—the only directly comparable prior result. Prior work reporting 95–99% accuracy [3–5] evaluates on isolated per-scenario sub-datasets, eliminating inter-class ambiguity and making those results not directly comparable.

5.2 RAG Retrieval Accuracy

The two-phase retrieval achieves perfect recall on all 13 curated protocol test cases: 13/13 (100%) of expected protocol identifiers appear in the retrieved context. Cross-encoder reranking is critical for several protocols (R4-PM, RTCI-01, FREQ-01) where the correct document is retrieved but does not rank first from the bi-encoder alone.

5.3 LLM Protocol Attribution and Ablation

Table 3: LLM protocol attribution accuracy and no-context ablation (same 200 samples, identical prompt structure). 95% Wilson CIs shown.

Condition	Correct	95% CI
Full pipeline (RAG context)	181/200 (90.5%)	[85.6%, 93.8%]
No-context ablation	33/200 (16.5%)	[12.0%, 22.3%]
χ^2 (p_0=1/13)		1931.42, $p < 10^{-10}$
Accuracy drop		−74.0 pp

The full AYLA pipeline achieves 90.5% correct protocol citation (181/200) with a 100% any-citation rate. A χ^2 goodness-of-fit test ($\chi^2 = 1931.42$, $p < 10^{-10}$) confirms performance far exceeds the random-selection baseline ($p_0 = 1/13 \approx 7.7\%$).

Removing retrieved context drops attribution to 16.5%—a **74.0 pp decline** with non-overlapping confidence intervals ([85.6%, 93.8%] vs. [12.0%, 22.3%]). Without context, the LLM collapses to citing a single default protocol (R4-PM) for nearly all events regardless of relay identity or sensor type, consistent with parametric guessing rather than document-grounded reasoning. This ablation directly confirms that the 90.5% pipeline accuracy is grounded in the retrieved protocol manuals.

6 Discussion

Remaining failure mode: RTCI-01. RTCI-01 (Remote Terminal Communication Injection) is the dominant unsolved failure class (0/7 = 0%). RTCI-01 involves current collapse correlated with Snort IDS log spikes; however, SHAP attribution frequently elevates phase-angle or current-harmonic sensors above the log channels when both co-occur, depriving the RAG query of the vocabulary needed to distinguish RTCI-01 from relay-specific FDIA scenarios. A log-channel promotion fix was attempted—elevating Snort log drivers in the RAG query regardless of SHAP rank when active—but this introduced new failures in other protocol classes: Snort log activity co-occurs across multiple attack types, so unconditionally promoting log channels over-retrieves RTCI-01 for events that are actually FDIA or FREQ-01. RTCI-01 remains an open failure class; resolving it likely requires a dedicated log-channel detection stage or protocol-aware retrieval filtering rather than a SHAP-rank override. FREQ-01 improved from 0% in the original single-LightGBM system to 75.0% with cross-encoder reranking, confirming that score incomparability across retrieval passes was the barrier, not missing protocol coverage.

Hardware and deployment. All results were produced on a laptop NVIDIA RTX 3060 (6 GB VRAM), averaging **3.5 minutes per event**. Because the LLM stage is event-driven—triggering only when a sensor snapshot exceeds τ, not on every relay log—the 3.5-minute cost is incurred only on flagged anomalies. For production: an RTX 4090 reduces latency to ∼45–90 s per event; an A100 to ∼20–40 s. For single-substation deployments with rare attack events, laptop-class hardware is viable; multi-substation control centers should use a dedicated GPU server. Local LLM deployment (Ollama) is required in environments where SCADA telemetry cannot be transmitted to external API endpoints.

7 Conclusion

Power Shield implements AYLA, an end-to-end pipeline for power grid attack detection and LLM-guided remediation. Three results establish its feasibility. **First**, the SHAP-guided LightGBM/XGBoost ensemble achieves 91.88% accuracy, AUC-ROC 0.9665, and 94.6% attack recall on the full combined MSU/ORNL dataset, surpassing the Hink et al. [8] combined-dataset baseline—the only directly comparable prior result. **Second**, a no-context ablation confirms that the 90.5% LLM protocol attribution (181/200) is grounded in retrieved documents: removing context drops accuracy to 16.5%, a 74.0 pp decline with non-overlapping 95% CIs. **Third**, unlike current commercial ICS security platforms (Claroty, Dragos, Nozomi Networks), which surface alerts but provide no automated remediation, AYLA delivers step-by-step, protocol-grounded operator guidance for each flagged event—a capability absent from all known deployed ICS security systems. The full implementation, pre-trained model, and evaluation data are publicly available at https://github.com/gradams42/power-shield.

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